Assignment 3: Sensor Characteristic

Feb 3, 2026

Small lecture:

- The bots sensors are so different than ours.
  - How you place and surround sensors greatly determines how well the robot will do
  - Sensors in body example
    - The human eye has a central point of focus but is within an eye ball that allow the eye to see different things. And also the cones are under a pupil that moves to control incoming light. Showing that the body are just as important as the sensor.

Potential Software logic

```
def Scan()
    motor swing full()
    degree = motor_degree
    light_max = [0, 0]
    if Sensor light ≥ light_max[0]
        light_max.replace [Sensor light, motor degree]
    return light max
```

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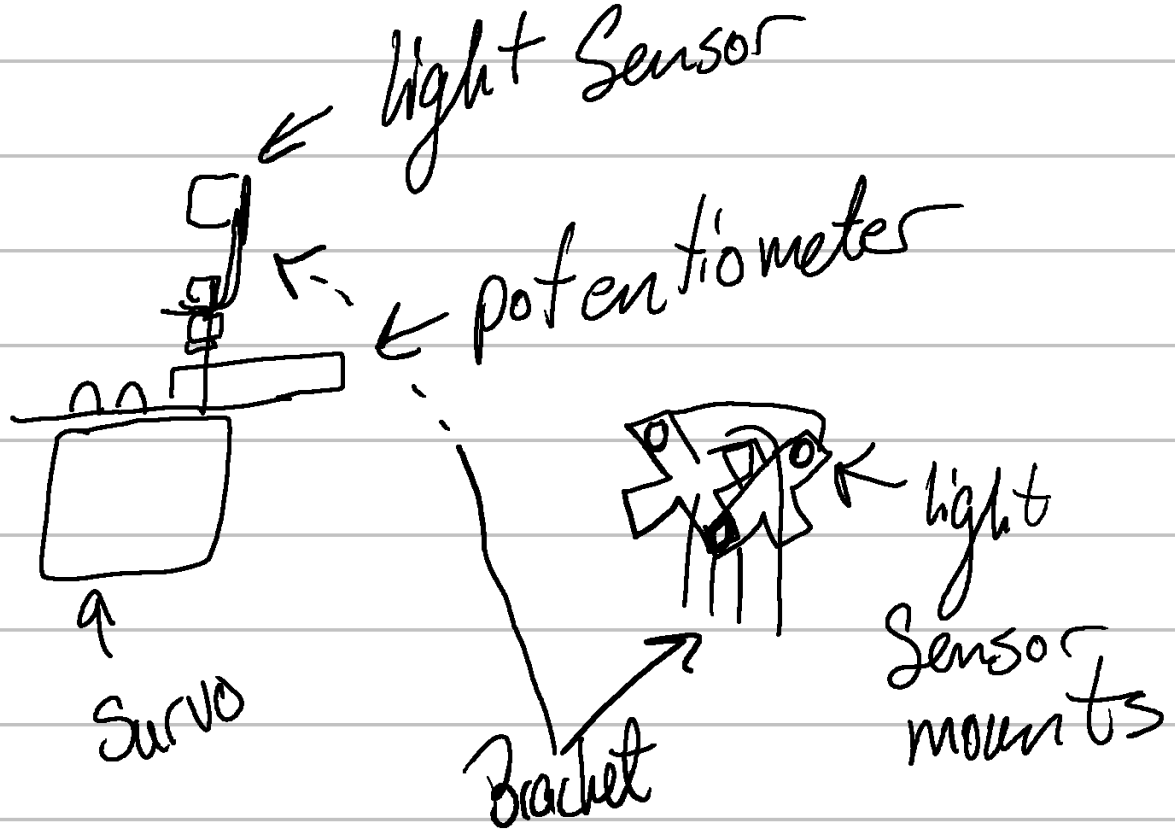
light Mount



Drive train we need to implement in order to allow the shrinking of code

Built the light Sensor Assembly

- A lot of the Axel Collars are stripped so we had

Code for the robot kept outputting errors

We attempted to make sure the sensor was centered.

PC angle : 114° - 214°

Got the dot to turn towards the light refrence to go

Part 1 complete

moved the time down to have it wait for less time

Part 2:

Measure various points of light source

Deg	Light	
-27° □	21.85331.1°	Close
-5° △	22.10404.1°	
14° 0	23.18615.1°	
-17° □	15.08353.1°	far
-3° △	14.687764.1°	
14° 0	16.00728	

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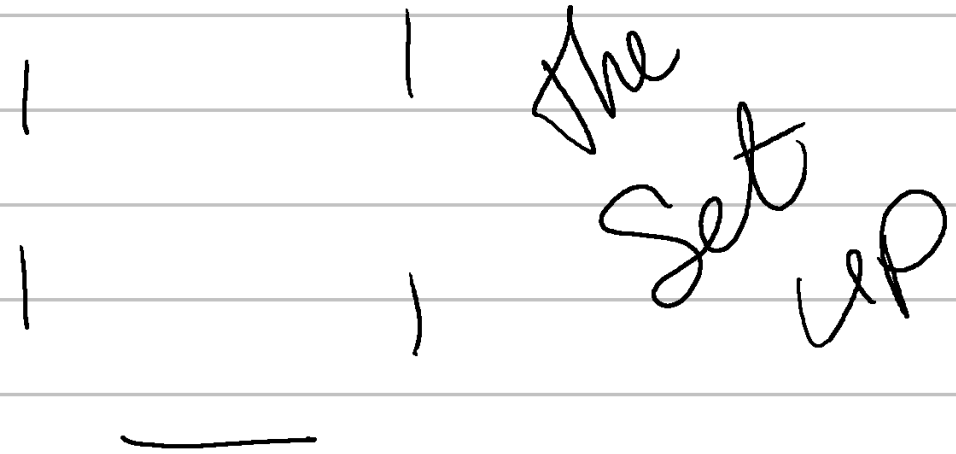
Potent Measure vs The light Sensor
degree



1. 186

2. 182

3. 163

4. 163

5. 143

6. 144

D	L
1. 193,	24.8489
2. 189,	16
3. 169,	22
4. 162,	15
5. 150,	15
6. 149,	14.7

No Blinder

1. $186 - 183 = 3 = 9$
2. $182 - 189 = -7 = 49$
3. $163 - 169 = -6 = 36$
4. $163 - 162 = 1 = 1$
5. $143 - 150 = -7 = 49$
6. $144 - 149 = -5 = 25$

$+ \; 169$

$169 \div 6 = \sqrt{28.17} = 5.307$

Small Blinder

1. $186 - 183 = 3^2 = 9$
2. $182 - 178 = 4^2 = 16$
3. $163 - 160 = 3^2 = 9$
4. $163 - 163 = 0^2 = 0$
5. $143 - 141 = 2^2 = 4$
6. $144 - 146 = -2^2 = 4$

$+ \; 42$

$42 \div 6 = \sqrt{7} = 2.65$

Big Blinder

1. $186 - 191 = -5^2 = 25$
2. $182 - 188 = -6^2 = 36$
3. $163 - 170 = -7^2 = 49$
4. $163 - 169 = -6^2 = 36$
5. $143 - 151 = -8^2 = 64$
6. $144 - 151 = -7^2 = 49$

$+$ $\Big\}$ 259

$259 \div 6 = \sqrt{43.17} = 6.57$

We have a theory we will turn down the speed

1. $186 - 190 = -4^2 = 16$
2. $182 - 189 = -7^2 = 49$
3. $163 - 169 = -6^2 = 36$
4. $163 - 172 = -9^2 = 81$ +
5. $143 - 151 = -8^2 = 64$
6. $144 - 151 = -7^2 = 49$

295

$295 \div 6 = \sqrt{49.1} = 7.167$

Result worse results when slowed down

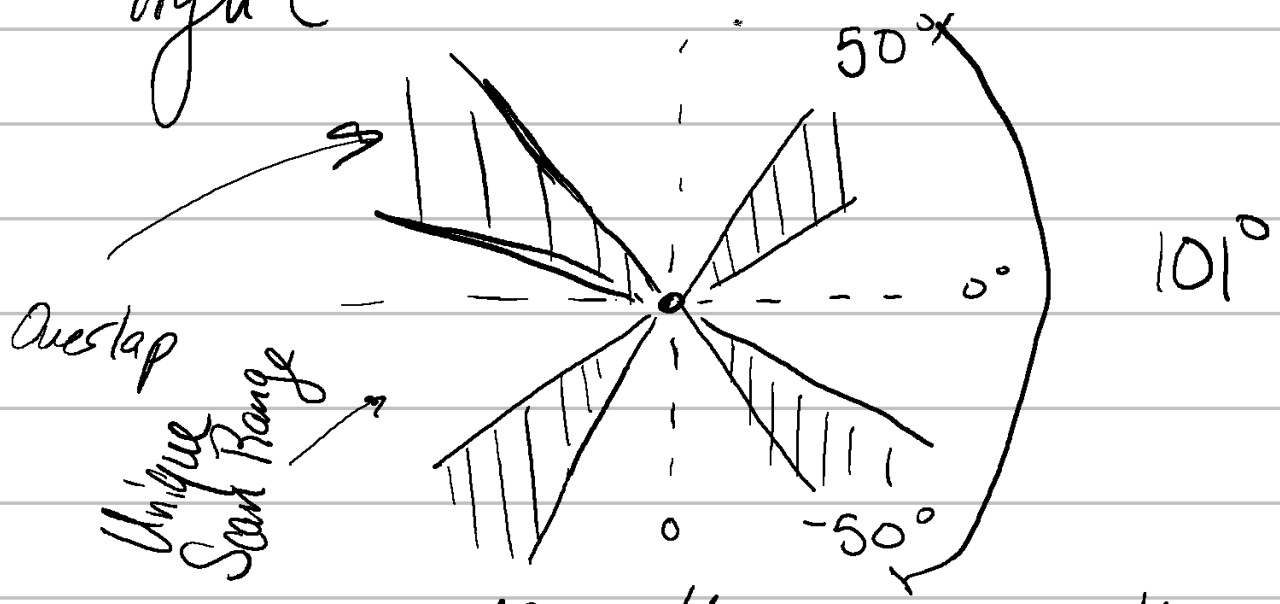
The Small Cone was the best overall error of 2.65.

Nothing: 5.307
Big Cone: 6.57

There is a noticible improvement in the small cone. we believe this is due to just how deep the large cone is that the sensor struggles to pick up the difference in light intensity.

Part 2 Complete Feb 6.

for the next part we need to have the bot drive towards the light



$101 \times 4 = 404°$ this means that the Bot will have over-lapping Scans ... good

Feb 10, 2026

We are going to have the bot scan all 4 directions constantly checking on whether the next scan produces a brighter light.

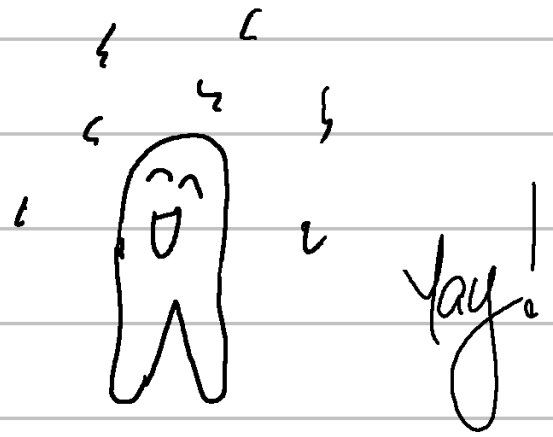
- We are going to keep the camera straight and make the bot spin 360° then run a scan.

Feb 12, 2026

We kept having an issue of an inf loop

- We over-engineered the code just a bit
- The issue is our scan code refuses to correctly scan light.

We got it to work



We had to inverse how the math computed the degree since we had to standardize it

We had an issue where the bot would see glare

The issue now is the light comparison finds the first bright and uses that angle instead

We fixed it by making it center the angle

angle + (count / 2)

2/13/20

Group completed the go to the light code

How they did that not a clue they don't remember

Something got changed

The code works but we need to implement a memory.

We have the values we have

> To solve the light sensing we made the range -30 → 30 as the glare from glossy obj. would register false profit

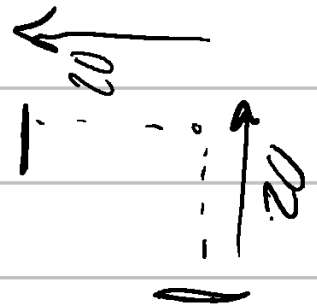
The bot is reading the Hue and will run a boolean comparison to check the color and see.

- Running into an issue in that on 90° turns the bot will turn infinitely.
- Put the 90° turn logic inside a while not loop

Checkpoint #1 ✓
Checkpoint #2 ✓

We now need to impliment the

Measure
X: 19.5 in.
X: 20 in.



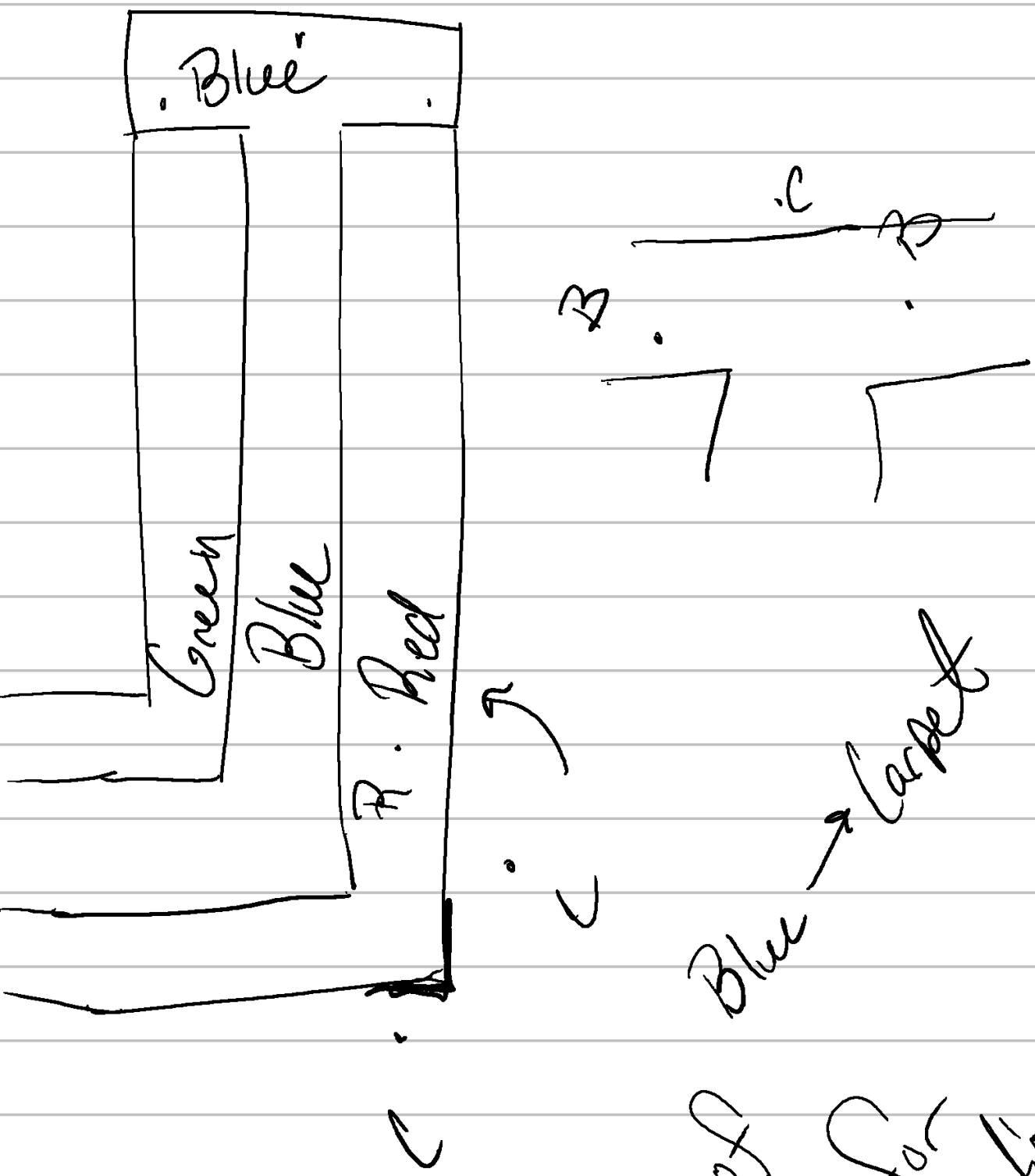


Diagram of Track for Conversation

We hard coded the turn distance seems to work
Issue currently is the heading

- Changed turn to be at 75°
Issue Heading isn't gurenteed
- We applyed a case to the blue → carpet where-in it will set its heading to 180°

- Issue Red/Green → Carpet will cause the Blue → Carpet func to break.

This works

all done.